

Dummit & Foote 7.4

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1 Problems

Let R be a commutative ring with identity, $A_1, A_2, \dots, A_n$ be ideals. The map $\varphi : R \rightarrow R/A_1 \times R/A_2 \times \dots \times R/A_n$ is a ring homomorphism with kernel $A_1 \cap A_2 \cap \dots \cap A_n$.

For simplicity take $n = 2$.

$$\begin{aligned}\varphi(r + s) &= (r + s + A_1, r + s + A_2) \\ &= (r + A_1, r + A_2) + (s + A_1, s + A_2) \\ &= \varphi(r) + \varphi(s).\end{aligned}$$

$$\begin{aligned}\varphi(rs) &= (rs + A_1, rs + A_2) \\ &= (r + A_1, r + A_2)(s + A_1, s + A_2) \\ &= \varphi(r)\varphi(s).\end{aligned}$$

$\varphi(r) = (r + A_1, r + A_2) = (A_1, A_2)$ means $r \in A_1$ and $r \in A_2$, or $r \in A_1 \cap A_2$. $\ker(\varphi) = A_1 \cap A_2$. Particularly when $A_1 \cap A_2 \cap \dots \cap A_n = 0$, φ is injective.

When $A_1, A_2, \dots, A_n$ are pairwise comaximal, then $R/(A_1 A_2 \dots A_n) = R/(A_1 \cap A_2 \cap \dots \cap A_n) \cong R/A_1 \times R/A_2 \times \dots \times R/A_n$.

When $n = 2$, $A_1 A_2 \subset A_1 \cap A_2$. Since $A_1 + A_2 = R$, there exist $a_1 \in A_1$ and $a_2 \in A_2$ such that $a_1 + a_2 = 1$. Let $x \in A_1 \cap A_2$. $x = a_1 x + x a_2 \in A_1 A_2$, so $A_1 A_2 = A_1 \cap A_2$. Consider $(r + A_1, s + A_2)$ in the codomain. There exist $a_1 \in A_1, a_2 \in A_2$ such that $a_1 - a_2 = s - r$, or $a_1 + r = a_2 + s = x$. Note $\varphi(x) = (r + A_1, s + A_2)$, so φ is surjective. The result for $n = 2$ follows from the First Isomorphism theorem.

Let $A_1, A_2, \dots, A_n$ be distinct pairwise comaximal ideals for $n > 1$. Then $A_1 A_2 \dots A_{n-1}, A_n$ are comaximal.

Proof: We induct on n . The result holds for $n = 2$ by definition of comaximal. Assume the result for $n = k \geq 2$ up to naming. Consider

$$\begin{aligned} R &= (A_1 A_2 \cdots A_{k-1} + A_{k+1})(A_k + A_{k+1}) \\ &= A_1 A_2 \cdots A_k + A_{k+1}(A_1 A_2 \cdots A_{k-1} + A_k + A_{k+1}) \\ &= A_1 A_2 \cdots A_k + A_{k+1}(R + A_{k+1}) \\ &= A_1 A_2 \cdots A_k + A_{k+1}. \end{aligned}$$

Hence, the result holds for $n = k + 1$. We now show φ is surjective by induction.

The result holds for $n = 2$ shown above. Suppose given $r_1, r_2, \dots, r_k$ there exists x_k such that $x_i + A_i = r_i + A_i$ for $i \in [k]$. Note $A = A_1 A_2 \cdots A_k, A_{k+1}$ are comaximal. There exists $a \in A$ and $a_{k+1} \in A_{k+1}$ such that $a - a_{k+1} = r_{k+1} - x_k$, or that $x_{k+1} = a + x_k = a_{k+1} + r_{k+1}$. Note that $x_i + A_i = r_i + A_i$ for all $i \in [k+1]$, completing the induction. φ is surjective, finishing the proof of CRT.

Textbook solution: for $n = 2$, we can show surjectivity by: let $x \in A, y \in B$ such that $x + y = 1$. $x \equiv 1 \pmod{B}$, $y \equiv 1 \pmod{A}$. $\varphi(r_2 x + r_1 y) = r_2(0, 1) + r_1(1, 0) = (r_1, r_2)$. In general we claim $A_1 A_2 \cdots A_k$ and A_{k+1} are comaximal. Let $x_i \in A_i, y_i \in A_{k+1}$ be such that $x_i + y_i = 1$. $1 = (x_1 + y_1)(x_2 + y_2) \cdots (x_k + y_k) \equiv x_1 x_2 \cdots x_k \pmod{A_{k+1}}$, so $1 \in A_1 A_2 \cdots A_k + A_{k+1}$ and A, A_{k+1} are necessarily comaximal.

As rings $\mathbb{Z}/n\mathbb{Z} \cong (\mathbb{Z}/p_1^{\alpha_1}\mathbb{Z}) \times (\mathbb{Z}/p_2^{\alpha_2}\mathbb{Z}) \times \cdots \times (\mathbb{Z}/p_n^{\alpha_n}\mathbb{Z})$. The corresponding groups of units are isomorphic as well.

The following questions assume R is commutative with identity $1 \neq 0$.

P 1. An element $e \in R$ is said to be idempotent if $e^2 = e$. Show that the sets eR and $(1 - e)R$ are two-sided ideals. Further $R \cong eR \times (1 - e)R$. Show that $e, 1 - e$ are identities of the subrings $eR, (1 - e)R$, respectively.

S 1. Consider in general aR for some $a \in R$. aR is closed under subtraction: $ar - bs = a(r - s) \in aR$. aR is closed under left, right multiplication, so is an ideal. Note that $er + (1 - e)r = r$, so $eR, (1 - e)R$ are comaximal. Also, $(eR)((1 - e)R) = e(1 - e)R = 0$, so $R \cong R/0 \cong eR \times (1 - e)R$ by CRT. Finally, $e(er) = er$ and $(1 - e)^2 r = (1 - 2e + e^2)r = (1 - e)r$, proving the last statement.

P 2. Let R be a finite boolean ring with identity $1 \neq 0$. Prove that $R \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \cdots \times \mathbb{Z}/2\mathbb{Z}$.

S 2. We use strong induction on $|R|$. When $|R| = 2$, the result holds. Otherwise pick $e \in R$ such that $e \neq 0, 1$. Note that $aR = 0$ means $a^2 = a = 0$. Hence, both eR and $(1 - e)R$ are nonzero rings. e is idempotent, so $R \cong eR \times (1 - e)R$, and by the IH applied to $eR, (1 - e)R$, R is a product of copies of $\mathbb{Z}/2\mathbb{Z}$.

P 3. Let R, S be rings with identities. Show that every ideal of $R \times S$ is of the form $I \times J$, where $I \triangleleft R$ and $J \triangleleft S$.

S 3. Let $K \triangleleft R \times S$. Let L, T be the set of left, right coordinates, respectively that appear in K . Note $K \subset L \times T$. Let $(a, b) \in L \times R$. There exist $(a, b') \in K$ such that $b' \in T$ and $(a', b) \in K$ such that $a' \in L$. Since $K \triangleleft R \times S$, $(a, b')(1, 0) = (a, 0) \in K$ and $(a', b)(0, 1) = (0, b) \in K$. Hence $(a, b) \in K$, so $K = L \times T$.

P 6. Let $f_1, f_2, \dots, f_k$ be integer polynomials of the same degree d and $n_1, n_2, \dots, n_k$ be pairwise coprime positive integers. Show that there exists a polynomial f of degree d such that $f \equiv f_i \pmod{n_i}$ for all i . Further when each f_i is monic, f can be chosen to be monic.

S 6. Let $n = n_1 n_2 \dots n_k$ and $n'_i = n/n_i$. Note that n'_i is relatively prime with n_i . Let t_i be an inverse of $n'_i \pmod{n_i}$. Define

$$f = t_1 n'_1 f_1 + t_2 n'_2 f_2 + \dots + t_k n'_k f_k.$$

Note that $f \equiv f_i \pmod{n_i}$ for all i . If f is zero or has degree less than d , increase t_1 so that the leading coefficient is nonzero.

We now prove the second part. t_1 can be replaced by $t_i + n_1 l$, so $f = f_0 + n l f_1$. The leading coefficient of f_0 is $t_1 n'_1 + t_2 n'_2 + \dots + t_k n'_k$, which is $1 \pmod{n}$. Hence, we may pick l such that f is monic.

P 7. Let $m|n$. Prove that the natural ring homomorphism $\mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ is also a surjection of the units: $(\mathbb{Z}/m\mathbb{Z})^\times \rightarrow (\mathbb{Z}/n\mathbb{Z})^\times$.

S 7. By CRT it suffices to show the result when m, n are prime powers. Consider $\mathbb{Z}/p^k\mathbb{Z} \rightarrow \mathbb{Z}/p^j\mathbb{Z}$, $k \geq j$. The units of both are the integers relatively prime to p , showing the result.